REPRESENTABLE OPEN IMMERSIONS ARE LIMIT PRESERVING ON OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. Reduction-type proof; all three inputs are named and their roles are displayed.

Allowed standard prerequisites (editorial).

- Objects-versus-morphisms equivalence. lemma-representable-by-spaces-limit-preserving (same chapter, named). Reduces the claim to local finite presentation of p.
- Property implication principle. algebraic-lemma-representable-transformations-property-implication (cross-chapter, named). Transfers local finite presentation from the representing fibre morphisms.
- Open immersions are lfp. spaces-morphisms-lemma-open-immersion-locally-finite-presentation (cross-chapter, named). Supplies the morphism-level fact.

Source reference. Original reference 045F: focused-seeds/geometry/S9-crit-open-immersion-limit-preserving/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06CX: sources/stacks-criteria/pinned-original-source.tex, source lines 661–661. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06CN: focused-seeds/geometry/S9-crit-open-immersion-limit-preserving/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $p : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$. Assume p is representable by algebraic spaces and an open immersion. Then p is limit preserving on objects.*

Proof. This follows from Lemma 06CX and (via the general principle Algebraic Stacks, Lemma 045F) from the fact that an open immersion of algebraic spaces is locally of finite presentation, see Morphisms of Spaces, Lemma 06CN. □

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.